# A Fully Integer, Subject-Independent sEMG Gesture Classifier ASIC: Design Decisions, Trade-offs and Verification

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## Abstract

We designed and verified a fixed-function digital core that classifies four hand and wrist gestures from 8 to 24 surface-EMG electrodes using only integer arithmetic and one $8 \times 8$ multiplier. Every design choice was settled by a measured, subject-independent experiment on the public FORS-EMG dataset (leave-one-subject-out, paired statistics). The choice of gesture set was the largest single factor (+14.1 pp). Classifier capacity did not matter: an MLP, logistic regression and LDA are statistically tied, and INT8 quantisation costs 0.18 pp on average. Replacing floating-point features with add, compare and shift operations and a shift normaliser costs no accuracy (77.79% vs 77.88%). The resulting INT8 LDA core is described in synthesisable Verilog. A bit-exact golden model reproduces the software accuracy (77.61% vs 77.83%, no detectable difference), and the RTL matches that golden model bit for bit at every channel count from 8 to 24, using at most 850 of the 1015 clock cycles available per sample. We also report the classifiers we evaluated and rejected (CNNs and trees).

## 1 Introduction

Wearable myoelectric control (prostheses, gesture interfaces) needs a classifier that runs continuously at microwatt power and works for a person it was never trained on. These two requirements drive every decision in this report.

- **Low power** favours integer arithmetic, few multiplications, small memories and a low clock rate (here 1 MHz).
- **Generalisation to a new user** is the hard part of sEMG classification: skin, fat, electrode placement and muscle physiology change the signal between people. We therefore report only subject-independent accuracy.

The project went through three phases: choosing the gesture set and classifier (Sections 3–5), making the front end integer-only (Section 6), designing and verifying the hardware (Sections 7–8). Section 9 collects every decision, the alternatives and the evidence in one table.

## 2 Dataset and evaluation protocol

**Data.** FORS-EMG: 19 subjects, 8 forearm electrodes, 985 Hz, 12 gestures in 3 forearm orientations, 5 trials each; every recording is 8000 samples (≈8.1 s).

**Signal chain.** A 20–450 Hz Butterworth band-pass and a 50 Hz notch ($Q = 30$). The filter is causal (forward only), because hardware cannot run the zero-phase filter used in the first experiments: zero-phase filtering applies $|H|^2$ and uses future samples.

**Windows.** 200 ms windows (196–197 samples) with a 100 ms stride.

**Split.** Leave-one-subject-out (LOSO): the held-out subject is never seen in training or normalisation. Within the training subjects, trial 5 is the validation set and trials 1–4 train. Windows are cut *after* the split. An early bug cut windows first; because neighbouring windows share half their samples, this leaked validation data into training and produced a false 63% ceiling that did not depend on model capacity.

**Statistics.** Accuracy is the mean over held-out subjects with a 95% confidence interval. Two methods are compared per subject (paired): mean difference, 95% CI of the difference, and the Wilcoxon signed-rank test. A difference is called real only if its CI excludes zero.

The repository copy of the dataset is missing Subject 1, so runs made for this report use 18 subjects; runs made on the full dataset use 19. Every table states which.

## 3 Gesture set

The first set, {Hand Close, Hand Open, Index, Index Little}, stopped at about 65% whatever the model (Table 1). Its confusion matrix showed that Index ↔ Index Little alone caused about 40% of all errors. These two finger movements recruit overlapping deep flexor compartments, which 8 surface electrodes cannot separate.

**Decision.** Use {Hand Close, Hand Open, Wrist Flexion, Wrist Extension}: gross hand and wrist movements driven by distinct, superficial muscles. This gained **+14.1 pp** (paired $t$-test $p = 3.8 \times 10^{-9}$), more than any later model or hardware choice. Larger vocabularies cost accuracy: 69.0% for the best 6 gestures and 48.8% for all 12 (Fig. 1). The remaining dominant confusion,

**Table 1:** INT8 LOSO accuracy by gesture vocabulary (19 subjects; best tied model).

| Gesture set | Accuracy | Chance | × chance |
|---|---|---|---|
| A: 4 incl. Index, Index Little | 65.3% | 25% | 2.6 |
| **B: 4 selected (final)** | **79.4%** | 25% | 3.2 |
| D: best 6 of 12 | 69.0% | 16.7% | 4.1 |
| C: all 12 | 48.8% | 8.3% | 5.9 |

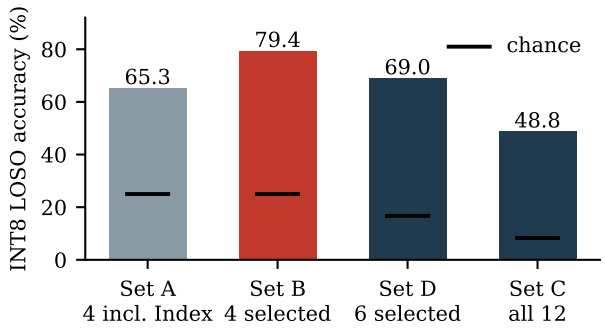



**Figure 1:** Accuracy for each gesture vocabulary; black ticks mark chance level.

Hand Open ↔ Wrist Extension, comes from shared extensor-digitorum activity.

The 6-gesture subset was chosen by ranking all $\binom{12}{6} = 924$ subsets by their worst pairwise confusion, using the 12-class confusion matrices. The predicted accuracy (66.4%, band 65–75%) contained the measured 69.0%, so existing confusion data can rank vocabularies without retraining.

# 4 Classifier

Eight classifiers were trained under one protocol on Gesture Set B, using 64 float features (8 types × 8 channels). The memory budget was 10 KB (Table 2, Fig. 2).

**Finding: capacity is not the bottleneck.** Five MLP widths (linear up to 64-128-4) were statistically indistinguishable on Set A. Logistic regression tied the MLP on the 4-, 6- and 12-gesture sets. The limit is the information in surface EMG, not the model.

**Decision: LDA for the chip.** Logistic regression had the best point estimate among the cheap models. We still chose LDA, for three reasons that matter in hardware:

1. its training is closed-form (class means and one shared covariance), with no iterative optimiser, so it can be retrained or adapted with a few matrix operations;
2. its weights are stored, not hard-wired, so a model retrained on the host is loaded through the configuration port without new silicon;
3. its scores have a probabilistic meaning (log-likelihoods up to a shared constant).

Its cost is 1.7 pp against the MLP, within the tie ($p = 0.09$). Trees were excluded: the accurate ones break the memory budget, and the only one inside it is significantly worse.

**Table 2:** Classifiers on Gesture Set B (INT8, LOSO, 19 subjects). "Tied": paired test against the MLP not significant.

| Model | INT8 % | MACs | Bytes | vs MLP |
|---|---|---|---|---|
| MLP 64-16-4 | 79.57 | 1088 | 1168 | – |
| Hist. grad. boosting | 79.26 | 0 | 19 200 | tied, >10 KB |
| Logistic regression | 79.19 | 256 | 272 | tied |
| Linear SVM | 78.30 | 256 | 272 | tied |
| LDA | 77.87 | 256 | 272 | tied ($p = 0.09$) |
| Random forest | 77.43 | 0 | 53 145 | worse, >10 KB |
| Extra trees | 76.04 | 0 | 38 015 | worse, >10 KB |
| Decision tree | 73.17 | 0 | 5055 | worse |

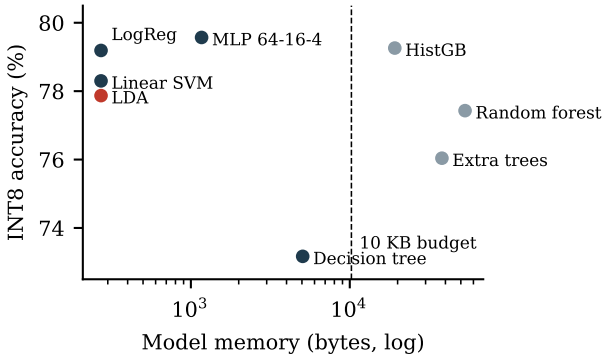



**Figure 2:** Accuracy against model memory. Grey: over the 10 KB budget. Red: the chosen LDA.

**CNNs and a programmable accelerator.** Yash and Digvijay trained raw-signal CNNs (depthwise-separable and two-branch designs, with batch normalisation folded into the weights and INT8 calibration) and co-designed an 8-bit ISA for a soft-programmable MAC accelerator. In October 2026 the hardware scope was fixed to the LDA:

- the classifier comparison showed no accuracy gain from capacity on hand-crafted features;
- a CNN needs many more MACs and weights;
- a programmable controller adds control logic and verification effort for a model the 4-gesture task does not need.

The CNN scripts remain in `analysis/` for comparison.

# 5 INT8 quantisation

Weights and activations are symmetric per-tensor INT8, with biases in INT32 on the same scale. Across 28 FP32-vs-INT8 comparisons (all models and gesture sets), the mean accuracy loss was **0.18 pp** (median 0.11, maximum 1.01 for the linear SVM). Twenty-five of the 28 losses were under 0.5 pp. A requantisation bug (an undefined shift when the multiplier exceeds 1) would have cost over 50 pp; a synthetic unit test caught it before any result was reported.

For a classifier that only needs arg max, requantisation is unnecessary. If every class score shares one positive scale $s > 0$, then

$$\arg\max_k s\, S_k = \arg\max_k S_k,$$

**Table 3:** Integer feature sets, INT8 LDA, LOSO (18 subjects). Δ: paired difference to R with the shift normaliser, 95% CI.

| | Power / ZC variant | Shift norm. % | Δ vs R (pp) |
|---|---|---|---|
| R | float reference | 77.45 (std) | – |
| H1 | exact $\sum x^2$ | 77.63 | $+0.19$ $[-0.36, +0.73]$ |
| H2 | Mitchell-2 | 77.56 | $+0.11$ $[-0.41, +0.63]$ |
| H3 | Mitchell-1 | 77.39 | $-0.05$ $[-0.57, +0.47]$ |
| H4 | $(\sum \lvert x\rvert)^2$ | 77.14 | $-0.31$ $[-0.88, +0.27]$ |
| H5 | no power term | 75.54 | $-1.91$ $[-2.94, -0.87]$ |
| **H6** | **Mitchell-1, ZC no dead-band** | **77.39** | $-0.05$ $[-0.59, +0.48]$ |

so the chip keeps one INT32 accumulator per class and never rescales. This requires a single weight scale for all classes.

# 6 Integer front end

The float features (MAV, WL, ZC, SSC, VAR) need division, square roots and floating-point thresholds. We rebuilt them from operations an ASIC does cheaply: add, subtract, absolute value, compare, shift, and a leading-one detector.

- **SABS** $= \sum |x|$ replaces MAV. The $1/N$ factor is absorbed by normalisation, since LDA is invariant to rescaling individual features.
- **WL** $= \sum |x_n - x_{n-1}|$.
- **ZC**: sign-bit XOR of consecutive samples, with no amplitude dead-band (`ZC_NODB`).
- **SSC**: strict sign change of consecutive differences; only the sign of the previous difference is stored, so each channel's history needs 10 bits.
- **POW**: $\sum x^2$ with Mitchell's one-step approximation of the square,

$$a^2 \approx a \cdot 2^{p+1} - 2^{2p}, \qquad p = \lfloor \log_2 a \rfloor,$$

  which is one leading-one detector, two shifts and one subtraction, never overestimates, and has correlation $r = 0.9999$ with the exact sum.

Log-amplitude features were rejected because a logarithm is expensive in hardware.

**Normaliser.** Z-scoring and float input quantisation are replaced by

$$q_f = \mathrm{clip}\big(((x_f - \mu_f) + 2^{k_f - 1}) \gg k_f,\ -128,\ 127\big),$$

with integer $\mu_f$ and $k_f$ calibrated on the training subjects.

**Result.** Six integer variants (H1–H6) were compared against the float reference on the same folds. Table 3 lists the results (18 subjects). With the integer normaliser, every variant that keeps a power feature is non-inferior to the float reference (margin $-1$ pp). Dropping the power term (H5) loses about 2 pp, so a power feature is required. Mitchell-1 squares (H3, H6) cost nothing measurable against exact squares.

**Decision**: H6, the cheapest variant, with the integer normaliser. On 19 subjects this fully integer front end gives 77.79±4.39% against 77.88% for the float pipeline.

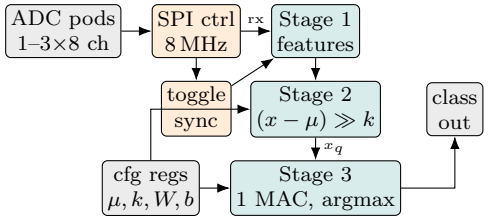



**Figure 3:** Chip partition. Orange: 8 MHz ADC domain. Teal: the 1 MHz datapath.

# 7 Hardware architecture

## 7.1 System partition

The question was how to support 8, 12, 16 or 24 electrodes. We considered three options: several compute boards exchanging partial LDA scores (exact, since the score is a sum over features), master/slave boards sharing state, or one compute core with pluggable acquisition pods. A compute-budget analysis settled it. At 1 MHz, features take about 25 cycles per channel per sample, so one core handles up to 40 channels within the 1000-cycle sample period, and the LDA needs $6(D + 1)$ cycles per decision. Distributing the computation saves nothing and adds synchronisation. **Decision**: one compute core plus 1–3 daisy-chained 8-channel ADC pods, with the channel count set by one register. Electrodes stay close to the muscle (short leads, less mains pickup), while all digital work stays on one chip.

## 7.2 Datapath

The datapath has three stages, all on `clk_sys` = 1 MHz (Fig. 3).

1. **Stage 1, features** (`emg_feat_accum`). For every frame of $C$ samples, a two-stage pipeline reads the channel's history and current bank word from RAM, computes the five features combinationally and writes the updated sums back: $C$ cycles per sample. Two ping-pong banks of 98 samples give the 196-sample window with a 98-sample stride and a single write pointer; a window is bank 0 + bank 1.
2. **Stage 2, normaliser** (`emg_shift_norm`). After every 98th sample, for each feature it adds the two bank fields, subtracts $\mu_f$, rounds, shifts by $k_f$ and clips: $D + 2$ cycles, with a 24-bit signed datapath derived from the field widths.
3. **Stage 3, LDA** (`emg_lda_mac`). One signed $8 \times 8$ multiplier and an INT32 accumulator. Per class, it loads the bias, then performs $D$ MACs, with one weight-address counter for all classes (so no class $\times D$ multiplication). A streaming argmax with strict $>$ (ties keep the lower class) replaces a comparator tree. Total $6(D + 1) + 2$ cycles; unused class slots get bias $-2^{31}$.

## 7.3 Clocking, crossing and reset

The ADC interface runs on an 8 MHz `clk_spi`. It generates CONVST every 8120 cycles (985 Hz), detects the

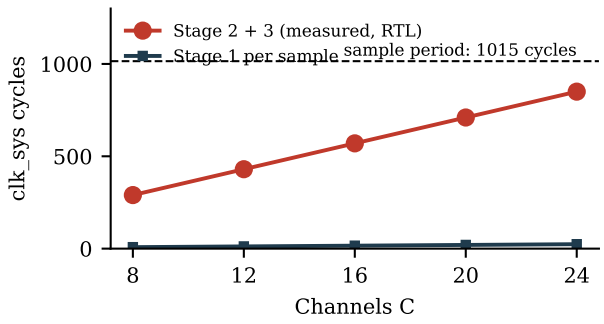



**Figure 4:** Cycle budget against channel count (RTL simulation, `clk_sys` = 1 MHz).

fall of DRDY through a 2-flip-flop synchroniser, and reads $8C$ bits at 4 MHz. Only one event crosses into the core: a frame toggle, through a 2-flip-flop toggle synchroniser. The data (`rx_flat`) crosses unsynchronised, because it is stable for about 1 ms around the event; in static timing analysis it is declared a multicycle path. Each domain has its own reset synchroniser (asynchronous assert, synchronous release). Constants and settings are loaded through one write port into writable memories, not ROM, because every channel count and electrode layout needs its own trained model.

### 7.4 Timing budget

Stage 1 costs $C$ cycles on every sample. Stages 2 and 3 run once per 98 samples, and their measured cost grows by 35 cycles per channel, from 290 to 850 (Fig. 4). Even at 24 channels, everything finishes within one 1015-cycle sample period. A single multiplier therefore suffices, and parallel MACs would buy nothing.

### 7.5 Memory

- feature history: $24 \times 10$ bits;
- feature banks: $2 \times 24 \times 64$ bits;
- $x_q$: $120 \times 8$ bits;
- normaliser constants: $120 \times 27$ bits;
- weights: $720 \times 8$ bits;
- biases: $6 \times 32$ bits.

The total is about 13.3 kbit, sized for the worst case of 24 channels and 6 classes. Every memory has a 1-cycle synchronous read, so the RTL maps to SRAM macros or to flip-flops unchanged.

## 8 Verification

**Golden model.** `lda_golden_model.py` reproduces every integer of Stages 1–3 in two forms: a sample-by-sample model that mirrors the RTL's counters, and a vectorised model used for training. The two are cross-checked on every run. Training uses the model's own bit-exact features, so training and hardware see identical numbers.

**Does the chip arithmetic cost accuracy?** The golden model (8-bit ADC, streamed 196-sample windows, shift normaliser, single weight scale) was compared with the software INT8 LDA on the same folds. Result: 77.61±4.34% against 77.83±4.11%, a difference of $-0.21$ pp with 95% CI $[-0.72, +0.30]$ and $p = 0.49$, so **no detectable difference** (18 subjects). The ranges actually used on real data leave wide margins:

- $\mu_f$: at most 13 of its 22 bits;
- $|S|$: at most 4047, against the INT32 limit.

**Unit tests.** The feature ALU was tested exhaustively on all 196 608 input combinations (every pair of INT8 values, each with the three signs of the previous difference). Each stage also had its own testbench. We checked that the testbenches catch real bugs by inserting deliberate ones: a logical instead of arithmetic shift, a missing rounding term, a wrong field, an unsigned multiply, $\geq$ instead of $>$, a dropped bias and a too-narrow accumulator. All were caught.

**Chip-level test.** A cocotb testbench (`RTL/tb/test_emg_top.py`) runs these steps:

1. it trains a model with the golden model and loads it through the configuration port;
2. it drives an ADC daisy-chain model with a gesture stream at the real clock rates;
3. at every stride it compares the $D$ values of $x_q$, the six class scores and the decided class against the golden model;
4. it checks that the first decision after reset is masked and that no overrun occurs.

A failure names the first stage that disagrees. The design passes at $C = 8, 12, 16, 20, 24$, bit-exactly, and lints cleanly in Icarus (`-Wall`) and Yosys.

## 9 Decisions and trade-offs

Table 4 lists every decision. Three trade-offs shaped the design most.

1. **Accuracy vs vocabulary.** Each added gesture costs accuracy. We favoured a reliable 4-gesture interface over a 12-gesture one at 49%.
2. **Model capacity vs cost.** A larger model (MLP, CNN) gains nothing measurable over the LDA but multiplies MACs and memory.
3. **Generality vs area.** One core with a channel-count register serves 8 to 24 electrodes; the cost is sizing every memory for 24 channels.

## 10 Conclusion and future work

Measured, subject-independent experiments justified each step. Gesture selection mattered most, model capacity did not, integer arithmetic and INT8 cost almost nothing, and a one-multiplier core at 1 MHz is enough for 24 electrodes. The RTL is verified bit-exactly against a model whose accuracy matches the software reference. Future work:

**Table 4:** Every major decision, the alternatives considered, and the evidence.

| Decision | Chosen | Alternatives | Reason / evidence |
|---|---|---|---|
| Evaluation | LOSO, trial-level validation, windows cut after the split | window-level split; per-subject training | only LOSO measures a new user; the window-level split leaked data (false 63% ceiling) |
| Gesture set | HC, HO, WF, WE | Index/Index Little set; 6 or 12 gestures | +14.1 pp; finger gestures inseparable with 8 surface electrodes; 6 gestures 69.0% |
| Classifier | INT8 LDA | MLP, LogReg, SVM, trees, CNN | all cheap linear models tied with the MLP; LDA has closed-form training and probabilistic scores; trees over budget or worse |
| Arithmetic | INT8 weights, INT32 accumulator, no requantisation | FP32; per-class scales | 0.18 pp mean loss; a shared scale keeps the argmax |
| Features | SABS, WL, ZC, SSC, Mitchell POW | float MAV/VAR/RMS, exact squares, log features | integer set matches the float reference; Mitchell square needs no multiplier |
| Normaliser | $(x - \mu) \gg k$, integer | z-score + float quantiser | no accuracy loss; only subtract, add and shift |
| Partition | 1 compute core + 1–3 ADC pods | multi-board partial sums; master/slave | one core handles up to 40 channels at 1 MHz; distributing adds synchronisation for no gain |
| Stage 3 | 1 MAC, streaming argmax | parallel MACs; comparator tree | 850 of 1015 cycles at $C = 24$; one MAC fits the budget |
| Windows | 2 ping-pong banks of 98 | sliding sums; 2 write pointers | one write pointer; window = bank0 + bank1 |
| Memories | writable RAM, 1-cycle read | ROM; flip-flops with combinational read | each channel count needs its own model; maps to SRAM or flip-flops |
| Clock crossing | toggle synchroniser for the event only | synchronise the data | data stable for about 1 ms; one bit crosses |
| Verification | bit-exact golden model, unit and chip tests, injected bugs | accuracy-only checks | every intermediate value checked; injected bugs caught |

1. synthesise in a target process to report area and power;
2. test the core with real ADC pods and recorded EMG.

# A Reproducing the results

All commands run from the repository root; each script prints the tables quoted above.

| Result | Script | Subjects |
|---|---|---|
| Gesture sets, classifiers, INT8 loss | `analysis/classifier_classification.py`, `K_4_model_MLP.py`, `K_6_model_LR_comparison.py` | 19 |
| Integer features H1–H6 | `analysis/K_4_LDA_HW_Experiments.py` | 18 |
| Feature-type selection | `analysis/fors_emg_feature_selection_lda.py` | 18 |
| Chip model vs software LDA | `golden_model/K_4_LDA_Golden_LOSO.py` | 18 |
| RTL | `RTL/tb/test_emg_top.py` (set `CHANNELS`) | synthetic |

# B Selected script outputs

Verbatim excerpts of the script printouts (18 subjects; the repository copy of the dataset lacks Subject 1). The full text is in `docs/report/outputs/script_outputs.txt`.

```
== golden_model/K_4_LDA_Golden_LOSO.py
--------------------------------------------------------------------------------
Model                    Test %
-----------------------  ------------
Chip LDA (golden model)  77.61 +-4.34
Validated INT8 LDA       77.83 +-4.11

Comparison           d pp  95% CI            Wilcoxon p  Up/Down    Verdict
-----------------  ------  --------------  ------------  ---------  ------------------------
chip vs validated   -0.21  [-0.72, +0.30]         0.486  9/9        no detectable difference

--------------------------------------------------------------------------------

== analysis/K_4_LDA_HW_Experiments.py
SABS vs float MAV              1

--------------------------------------------------------------------------------
  Accuracy (mean over 18 held-out subjects; +- = 95% CI)
--------------------------------------------------------------------------------
    Configuration                    Train INT8 %    Test FP32 %  Test INT8 % (std norm)    Test INT8 % (shift norm)
--  -------------------------------  --------------  -------------  ------------------------  --------------------------
R   Float reference                          81.56          77.27  77.45 +-4.89
H1  Exact Sum x^2 (serial shift-add)          81.51          77.2   77.33 +-4.89              77.63 +-4.67
H2  Mitchell-2 Sum x^2                        81.52          77.2   77.29 +-4.93              77.56 +-4.70
H3  Mitchell-1 Sum x^2                        81.5           77.2   77.26 +-4.92              77.39 +-4.68
H4  (Sum|x|)^2 per window                     81.34          76.96  77.10 +-4.95              77.14 +-4.72
H5  No power term                            79.41          75.43  75.38 +-5.16              75.54 +-5.06
H6  ZC no deadband + Mitchell-1              81.51          77.21  77.29 +-4.89              77.39 +-4.69

--------------------------------------------------------------------------------
  Paired comparisons, test INT8 (non-inferiority margin -1 pp)
--------------------------------------------------------------------------------
                Question                        d pp  95% CI            Wilcoxon p  Up/Down    NI
--------------  ------------------------------  ------  --------------  ------------  ---------  ----
H1 vs R         integer features (std norm)     -0.12  [-0.27, +0.03]         0.039  5/12       PASS
H2 vs R         integer features (std norm)     -0.16  [-0.30, -0.01]         0.028  6/12       PASS

== analysis/fors_emg_feature_selection_lda.py (stage 4)
--------------------------------------------------------------------------
Top-10 subsets                      k  Inner acc      Deficit pp    paired SE pp
---------------------------------  ---  -----------  ------------  --------------  ----------
{MAV, WL, ZC, SSC, VAR}              5  77.19%                 0               0     <- selected
{MAV, WL, ZC, SSC, VAR, WAMP}        6  77.10%                 0.09            0.08
{MAV, RMS, WL, ZC, SSC, VAR}         6  77.09%                 0.11            0.14
{MAV, RMS, WL, ZC, SSC, VAR, WAMP}   7  77.04%                 0.15            0.17
{MAV, WL, ZC, VAR}                   4  76.77%                 0.42            0.25
{MAV, RMS, WL, ZC, VAR}              5  76.74%                 0.45            0.23
{MAV, WL, ZC, VAR, WAMP}             5  76.72%                 0.47            0.25
{MAV, RMS, WL, ZC, VAR, WAMP}        6  76.66%                 0.53            0.24
{RMS, WL, ZC, SSC, VAR}              5  76.55%                 0.65            0.24
{RMS, WL, ZC, SSC, VAR, WAMP}        6  76.53%                 0.67            0.25

  * Final feature types : {MAV, WL, ZC, SSC, VAR}  (k = 5 of 8)
    Input dim           : 40  (was 64)
    LDA param bytes     : 176 (was 272)   MACs: 160 (was 256)
    Expected accuracy on a new subject (nested INT8): 77.08 +-4.97 %
```